NYC Airbnb EDA Report

Business Questions, Charts, Answers, and Recommendations

Project Type: Exploratory Data Analysis | Tooling: Python, Pandas, Seaborn, Matplotlib

Executive Summary

This report translates the Python EDA notebook into a recruiter-friendly business story. Each section connects a practical business question to a chart, the answer from the data, and an action-oriented recommendation.

- Brooklyn has the highest listing supply with 7,295 listings, representing 37.72% of focused listings.
- Manhattan has the highest median nightly price at $140, followed by Brooklyn at $120.
- Greenwich Village and West Village are the top premium neighborhoods among areas with at least 50 listings, both with a $200 median nightly price.
- Entire home/apt is the dominant room type, representing about 53.7% of focused listings.
- High-availability listings represent 52.29% of the market, indicating a large pool of inventory that may need pricing or demand optimization.
- Single-listing hosts are the largest host segment with 43.19% of listings, but professional hosts still represent a meaningful share of supply.

Dataset Snapshot

Metric	Value
Total listings analyzed	19,342
Median nightly price	$120
Average nightly price	$136
Median availability days	209
Average reviews per listing	43.6
Unique neighborhoods	221
Unique hosts	11,620

Methodology

- Cleaned and standardized column names to make the analysis reproducible and readable.
- Removed duplicate records and converted identifier fields into strings to avoid treating IDs as numeric measures.
- Parsed date fields and converted rating, bedroom, bed, and bath fields into numeric variables where needed.
- Used a focused analysis dataframe by removing extreme price outliers using an IQR-based threshold, while preserving the cleaned base data.
- Created business features including price per bed, availability bucket, host portfolio type, stay policy, review recency, and estimated revenue proxy.
- Used median price as the primary pricing metric because nightly Airbnb prices are skewed and outlier-sensitive.

Business Question 1: Where is Airbnb supply concentrated?

Why this matters: Supply concentration identifies competitive markets and shows where hosts are most likely to face pricing pressure.

Figure 1: Listing supply by borough.

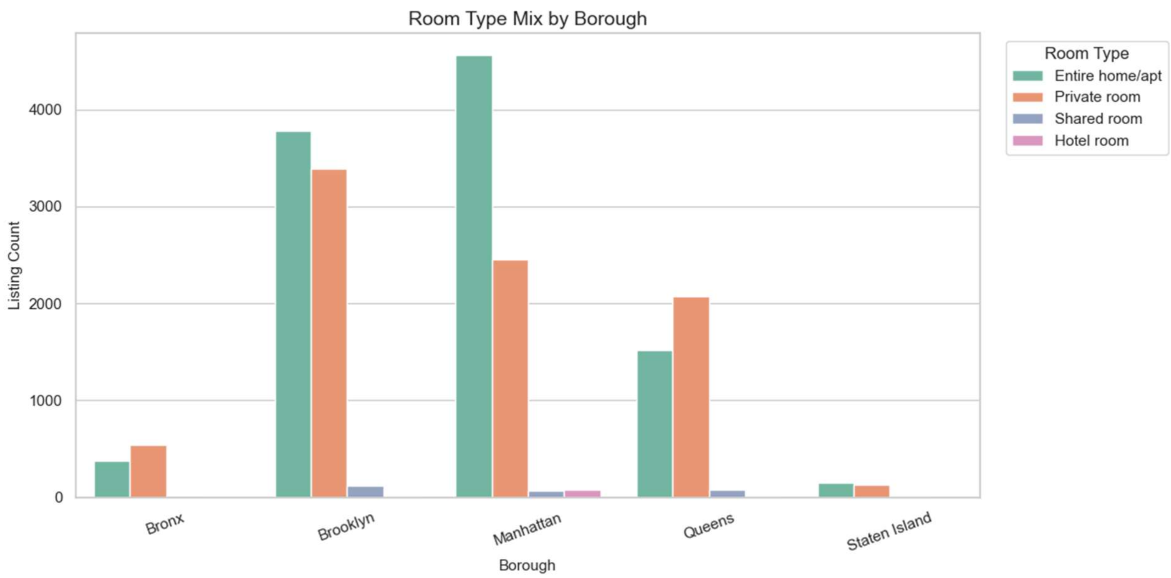

Figure 2: Room type mix by borough.

Answer from the analysis:

- Brooklyn leads supply with 7,295 listings, representing 37.72% of the focused dataset.
- Manhattan is nearly tied with 7,157 listings, representing 37.00% of the focused dataset.
- Brooklyn and Manhattan together account for 74.72% of focused listings, making them the core competitive markets.
- Queens is the next largest market with 3,670 listings, while Bronx and Staten Island have much smaller inventory shares.
- Entire home/apt is the most common room type, representing about 53.7% of focused listings.

Business takeaway: Market strategy should prioritize Brooklyn and Manhattan first because they dominate supply. However, pricing decisions should not use borough alone; room type must be included because the inventory mix differs meaningfully across boroughs.

Supporting Table: Borough Supply

Borough	Listings	Listing Share
Brooklyn	7,295	37.72%
Manhattan	7,157	37.00%
Queens	3,670	18.97%
Bronx	934	4.83%
Staten Island	286	1.48%

Business Question 2: How does pricing vary by borough and room type?

Why this matters: Pricing analysis helps hosts benchmark listings against the right market segment instead of comparing against all listings at once.

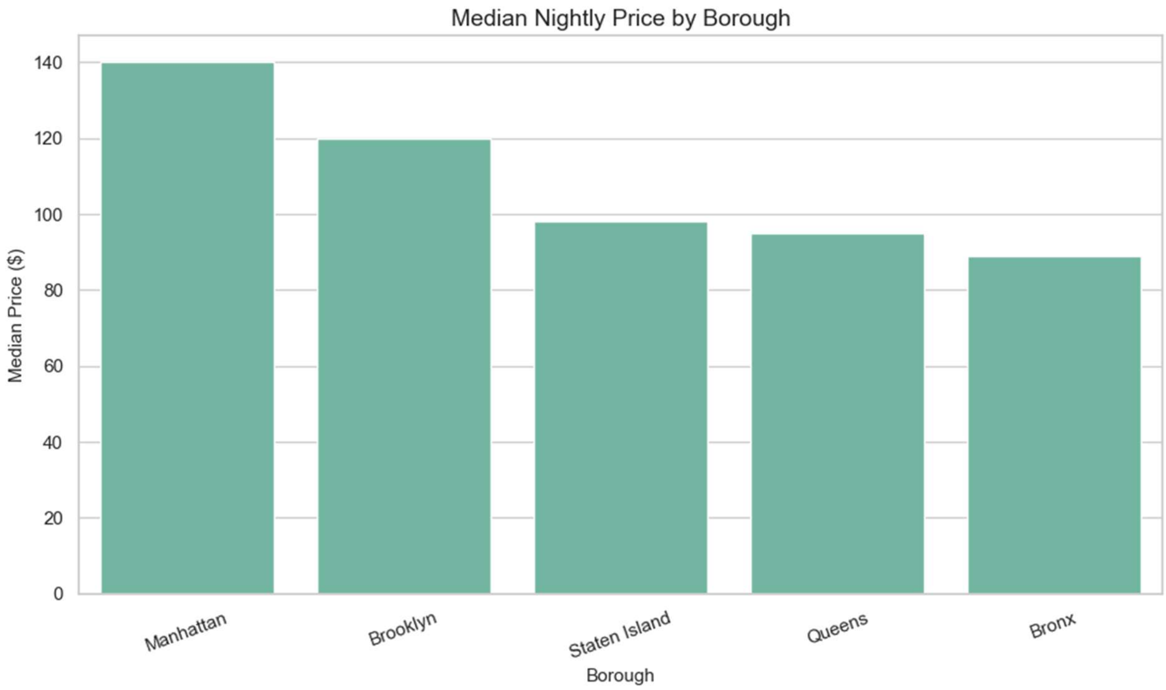

Figure 3: Median nightly price by borough.

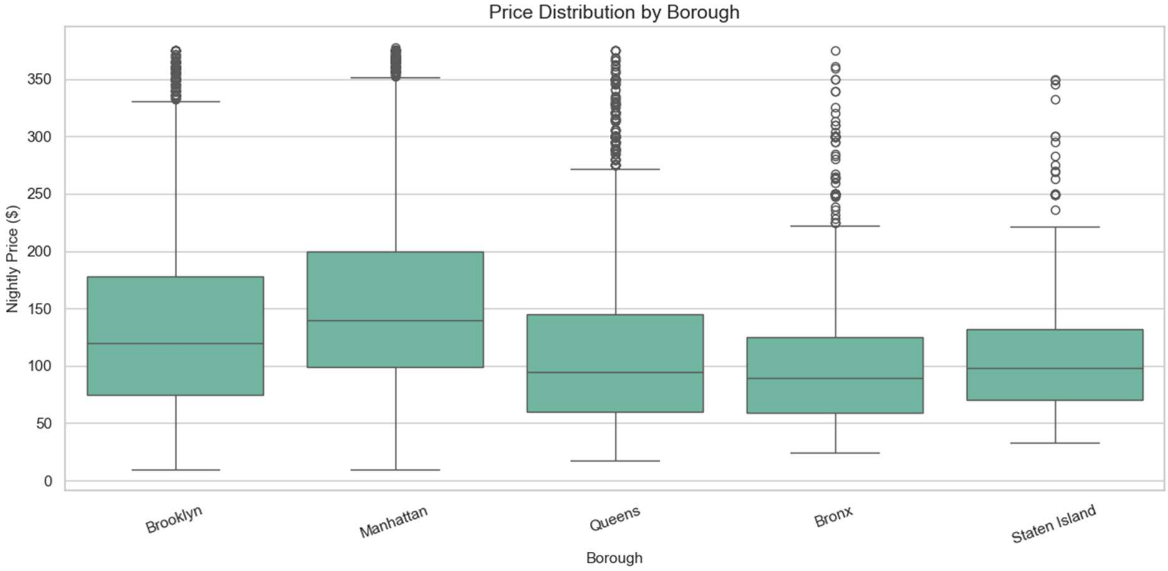

Figure 4: Price distribution by borough.

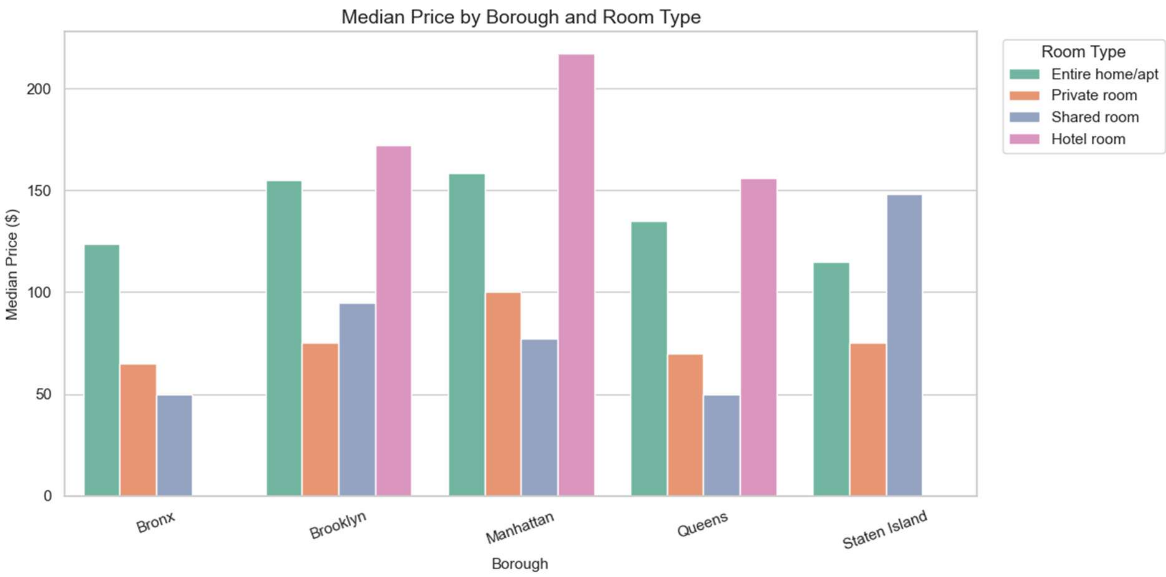

Figure 5: Median price by borough and room type.

Answer from the analysis:

- Manhattan is the highest-priced borough with a median nightly price of $140 and average price of $155.75.
- Brooklyn is the second-highest market with a median price of $120 and average price of $133.89.
- Bronx has the lowest borough-level median price at $89.
- Room type creates major pricing separation: entire homes and hotel rooms generally command higher prices than private or shared rooms.
- Manhattan hotel rooms show the highest segment median at $217, while Manhattan entire homes have a median of $158.50.

Business takeaway: Pricing recommendations should be segmented by borough and room type. A Manhattan entire home should not be benchmarked against a Queens private room or a Bronx shared room.

Supporting Table: Pricing by Borough

Borough	Listings	Median Price	Average Price	Median Price/Bed
Manhattan	7,157	$140	$155.75	$100.00
Brooklyn	7,295	$120	$133.89	$77.33
Staten Island	286	$98	$110.58	$60.00
Queens	3,670	$95	$112.46	$65.00
Bronx	934	$89	$102.18	$62.50

Business Question 3: Which neighborhoods have premium pricing?

Why this matters: Neighborhood-level pricing prevents broad borough averages from hiding high-value micro-markets.

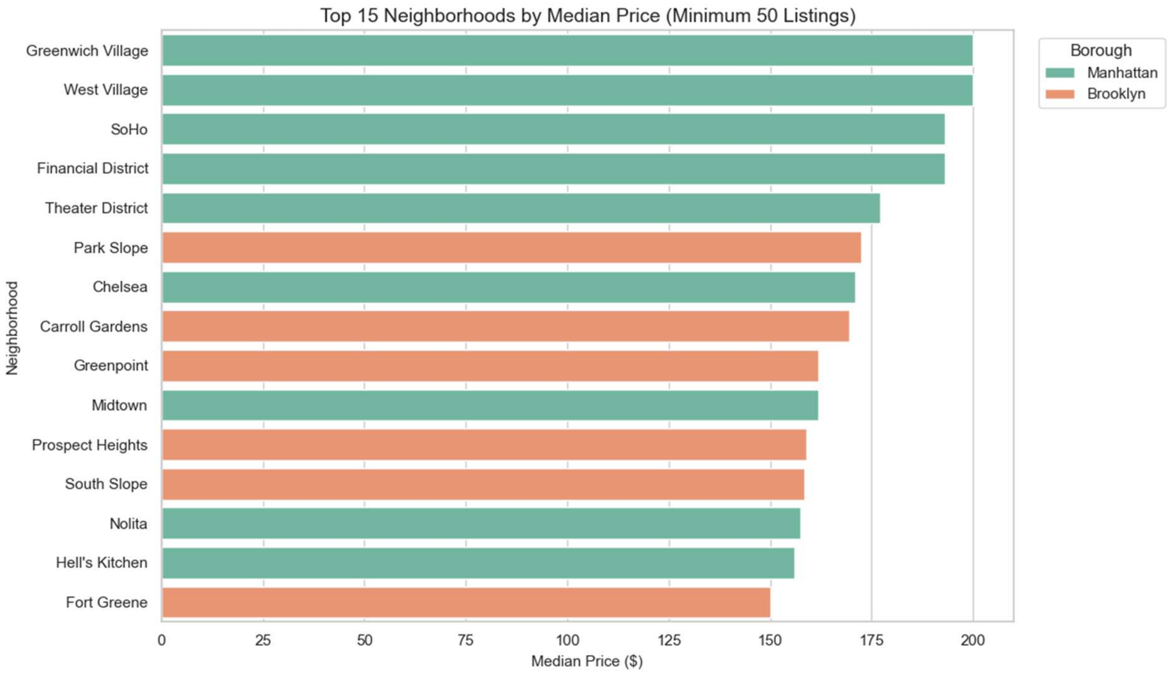

Figure 6: Top 15 neighborhoods by median price, using a minimum listing threshold of 50.

Answer from the analysis:

- Greenwich Village and West Village are the top premium neighborhoods with median nightly prices of $200.
- SoHo and Financial District follow closely, both with median prices of $193.
- Most of the top premium neighborhoods are in Manhattan, but Brooklyn has strong premium pockets such as Park Slope, Carroll Gardens, Greenpoint, Prospect Heights, South Slope, and Fort Greene.
- Midtown has high supply with 816 listings and a median price of $162, making it a large premium market rather than a small niche segment.

Business takeaway: For portfolio planning, premium opportunity should be evaluated at the neighborhood level. Manhattan dominates premium pricing, but selected Brooklyn neighborhoods also offer strong price positioning.

Supporting Table: Premium Neighborhoods

Neighborhood	Borough	Listings	Median Price	Average Price	Median Availability
Greenwich Village	Manhattan	69	$200	$209.52	173
West Village	Manhattan	171	$200	$206.71	127
SoHo	Manhattan	105	$193	$208.82	177
Financial District	Manhattan	163	$193	$192.53	173

Theater District	Manhattan	118	$177	$191.62	281
Park Slope	Brooklyn	144	$172.50	$171.47	165
Chelsea	Manhattan	309	$171	$183.31	242
Carroll Gardens	Brooklyn	70	$169.50	$178.41	178
Greenpoint	Brooklyn	275	$162	$169.27	180
Midtown	Manhattan	816	$162	$173.59	179

Business Question 4: How do availability and stay policies affect the market?

Why this matters: Availability and minimum-night rules help identify whether listings are easy to book, underutilized, or restricted by policy choices.

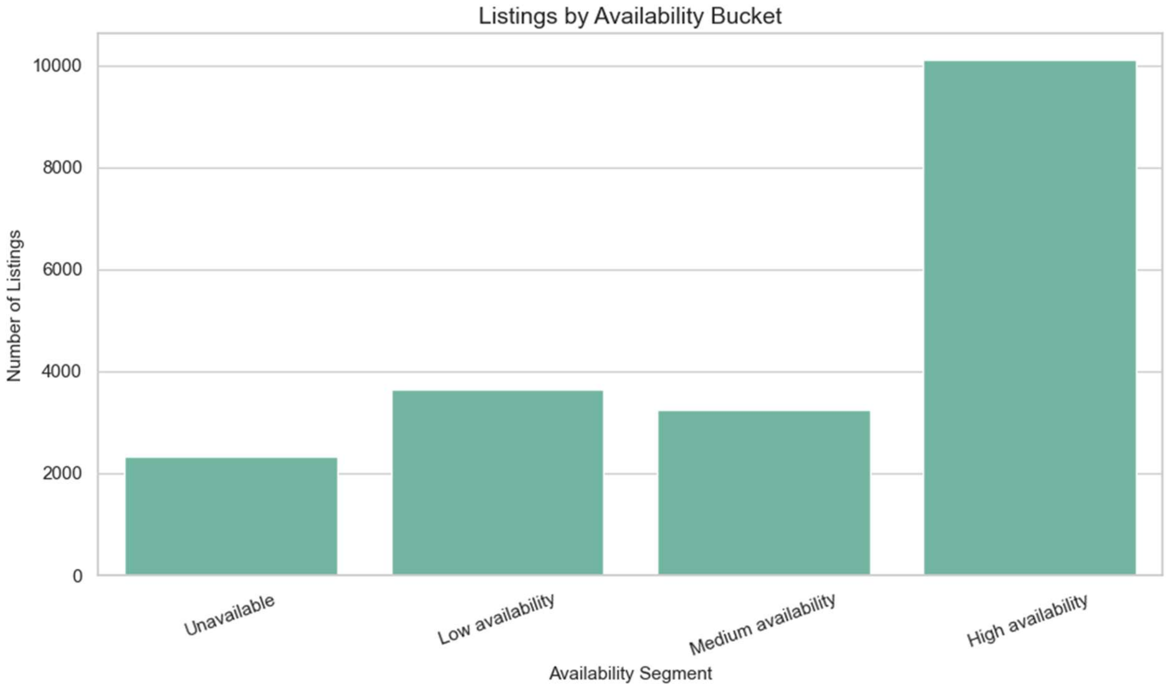

Figure 7: Listings by availability bucket.

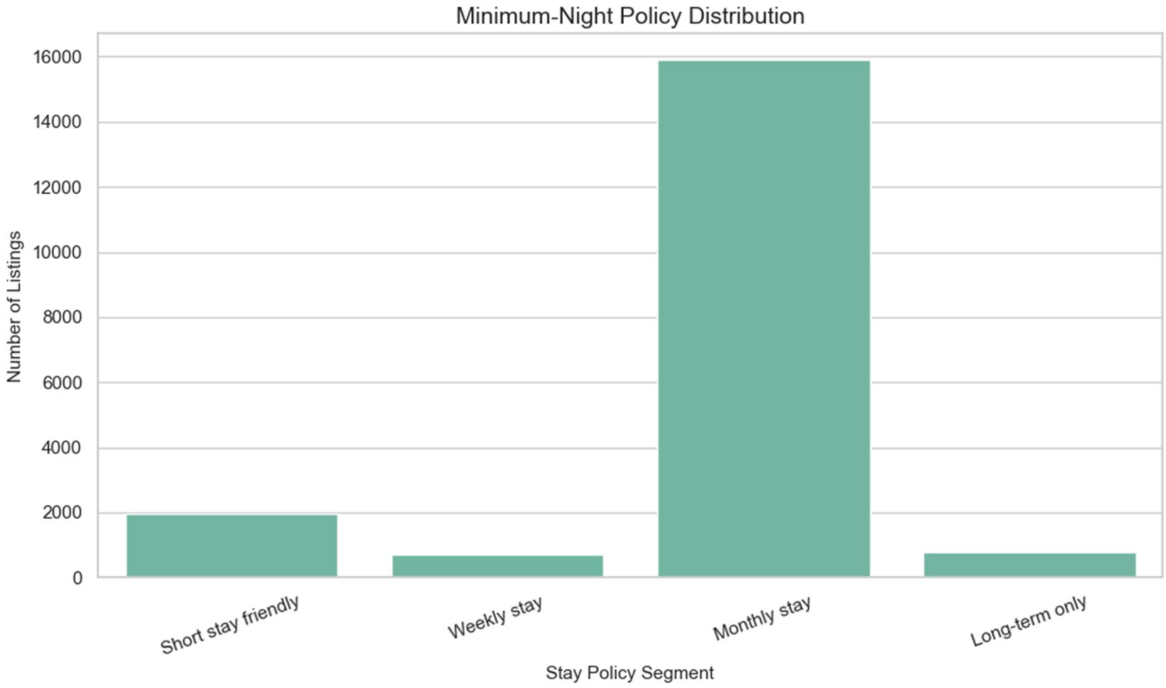

Figure 8: Minimum-night policy distribution.

Answer from the analysis:

- High-availability listings account for 10,113 listings, or 52.29% of the focused dataset.
- Low-availability listings account for 18.88%, while unavailable listings account for 12.04%.
- Monthly stay policies dominate the market with 15,909 listings, representing 82.25% of focused listings.
- Short stay friendly listings have a higher median price of $153, and weekly stay listings have the highest median price at $170.
- Monthly stay listings have a lower median price of $111, suggesting that restrictive minimum-night policies may limit price potential or reflect a different customer segment.

Business takeaway: Listings with high availability should be reviewed for pricing, positioning, and minimum-night rules. Shorter stay flexibility appears connected with stronger median pricing, but the business should validate this with booking and occupancy data before changing policy.

Supporting Table: Availability Segments

Availability Segment	Listings	Median Price	Listing Share
Unavailable	2,328	$120	12.04%
Low availability	3,651	$110	18.88%
Medium availability	3,250	$120	16.80%
High availability	10,113	$120	52.29%

Supporting Table: Stay Policy Segments

Stay Policy	Listings	Median Price	Listing Share
Short stay friendly	1,947	$153	10.07%
Weekly stay	716	$170	3.70%
Monthly stay	15,909	$111	82.25%
Long-term only	770	$125	3.98%

Business Question 5: What demand signals appear in reviews?

Why this matters: Reviews are not a perfect demand metric, but review volume, review recency, and reviews per month are useful engagement signals in marketplace analysis.

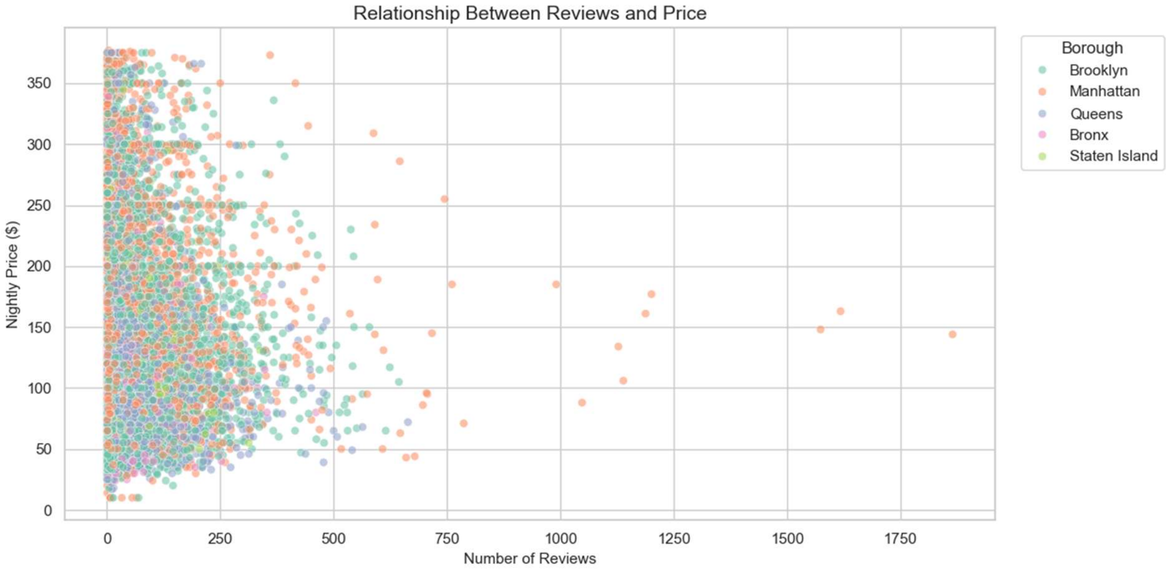

Figure 9: Relationship between review count and nightly price.

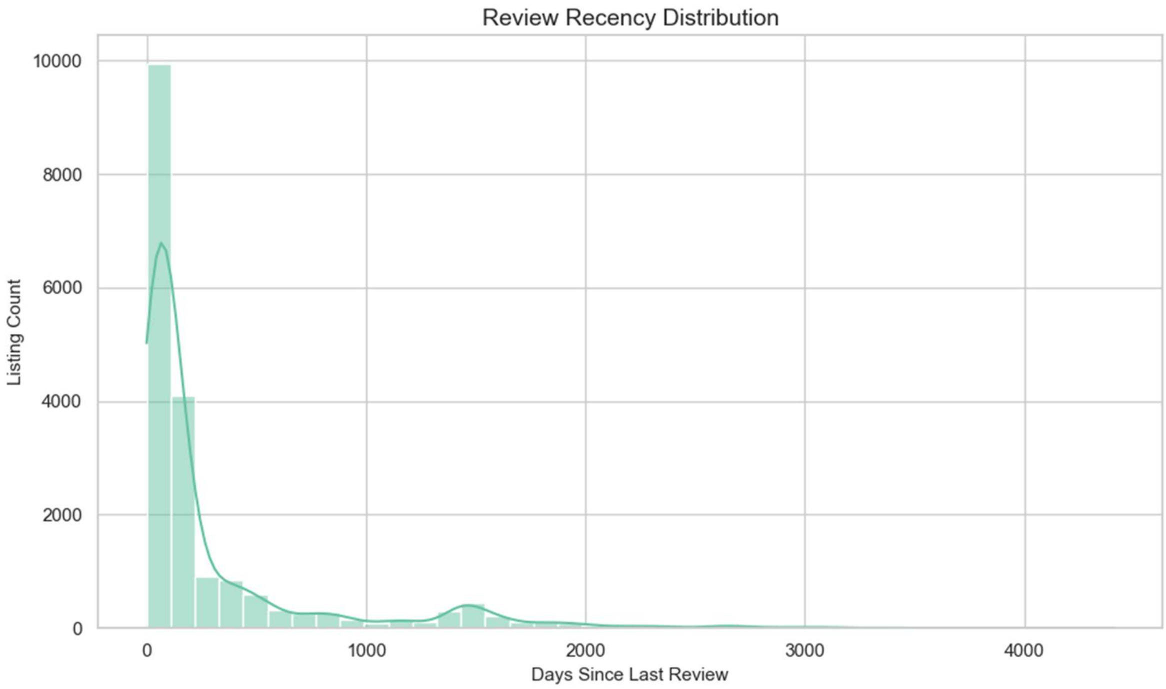

Figure 10: Review recency distribution.

Answer from the analysis:

- Review activity exists across a wide range of prices, so higher price alone does not guarantee stronger engagement.
- Manhattan private rooms show strong review velocity with about 1.54 average reviews per month and a median price of $100.

- Queens entire homes also show strong review velocity with about 1.53 average reviews per month and a median price of $135.
- Manhattan entire homes have a higher median price of $158.50 but lower review velocity at about 0.95 average reviews per month.
- The review recency distribution helps flag listings that may be stale, inactive, or less competitive in recent demand.

Business takeaway: Hosts should not only optimize for price; they should monitor review velocity and recent guest activity. A listing with high price but weak recent reviews may need repositioning, pricing adjustments, or listing quality improvements.

Business Question 6: Are listings driven by individual or professional hosts?

Why this matters: Host segmentation reveals whether market behavior is shaped by casual individual hosts or operators managing multiple listings.

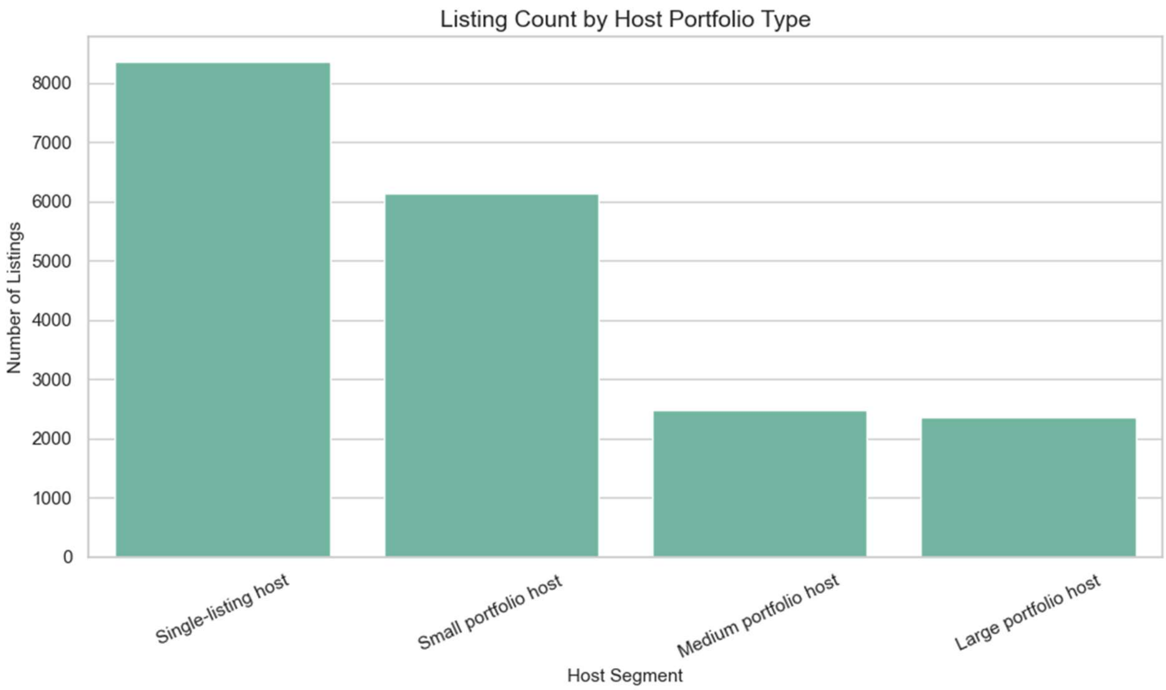

Figure 11: Listing count by host portfolio type.

Answer from the analysis:

- Single-listing hosts are the largest segment with 8,354 listings, representing 43.19% of focused listings.
- Small portfolio hosts account for 6,144 listings, representing 31.77% of the focused dataset.
- Medium and large portfolio hosts together account for 4,844 listings, or about 25.04% of focused listings.
- Large portfolio hosts have the highest median availability at 281 days, which may suggest more professionally managed inventory or weaker booking velocity in some portfolios.
- Urban Furnished is the top host by listing count with 146 listings, while Blueground has a high median price of $266 among the top operators.

Business takeaway: Host strategy should separate individual hosts from portfolio operators. Professional hosts may require different pricing, onboarding, compliance, and revenue management strategies than single-listing hosts.

Supporting Table: Host Portfolio Segments

Host Segment	Listings	Unique Hosts	Median Price	Median Availability	Listing Share
Single-listing host	8,354	8,354	$135	177	43.19%
Small portfolio host	6,144	2,815	$102	233	31.77%
Medium portfolio host	2,488	381	$102	269	12.86%
Large portfolio host	2,356	70	$117	281	12.18%

Supporting Table: Top Hosts by Listing Count

Host Name	Listings	Median Price	Average Availability
Urban Furnished	146	$139	314.85
Jeniffer	129	$120	303.61
Stay With Vibe	113	$99	295.56
Blueground	96	$266	288.29
Hiroki	87	$38	48.07

Business Question 7: Where are listings geographically clustered?

Why this matters: Spatial analysis helps identify listing density, neighborhood clusters, and geographic patterns that table summaries may miss.

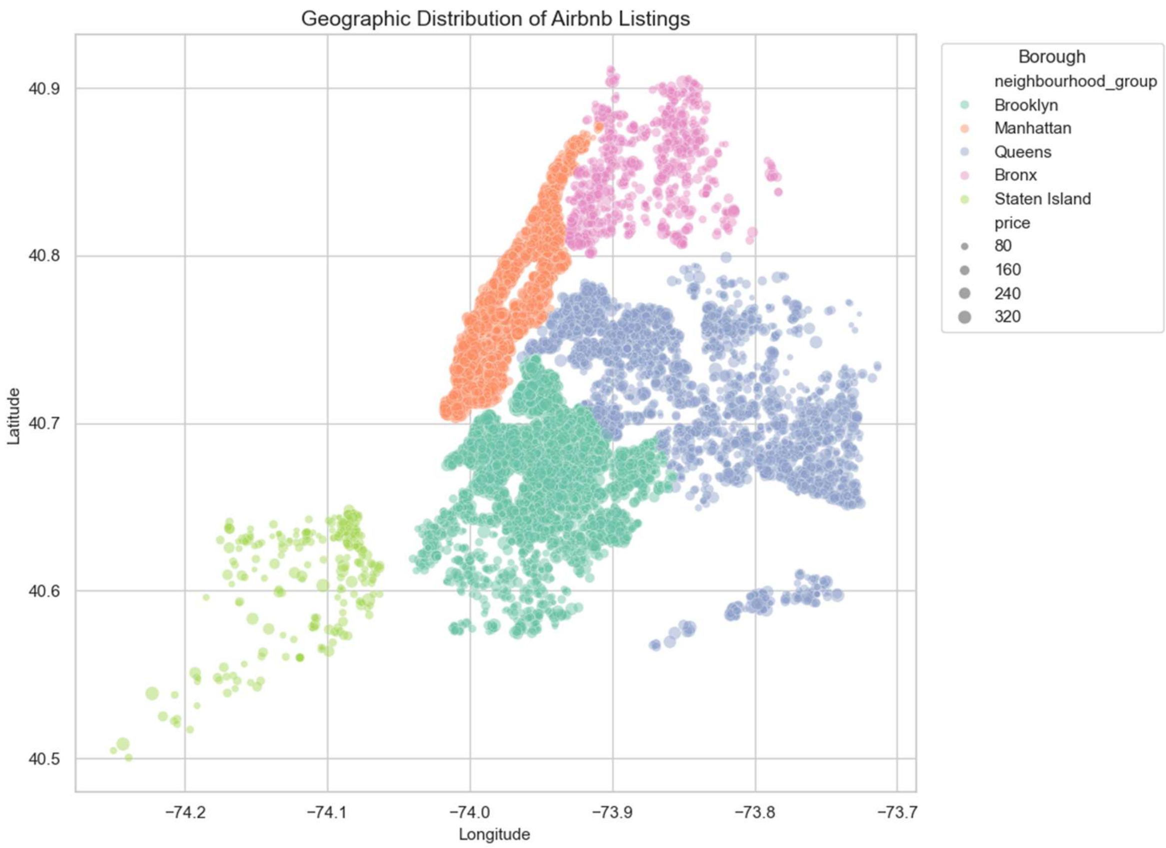

Figure 12: Geographic distribution of Airbnb listings.

Answer from the analysis:

- Listings are heavily clustered in the Manhattan, Brooklyn, and Queens urban core.

- Bronx and Staten Island have visibly lower listing density, consistent with their smaller supply shares.
- Higher-priced listings appear more concentrated in central, dense neighborhoods, especially Manhattan and selected Brooklyn areas.

Business takeaway: Geographic segmentation can support neighborhood expansion planning, localized pricing, and supply monitoring. Spatial features should be considered in any future predictive model.

Business Question 8: Which listings may represent pricing or revenue opportunities?

Why this matters: Combining price and availability creates a practical opportunity framework for identifying listings that may need pricing action.

Figure 13: Opportunity segments based on price and availability.

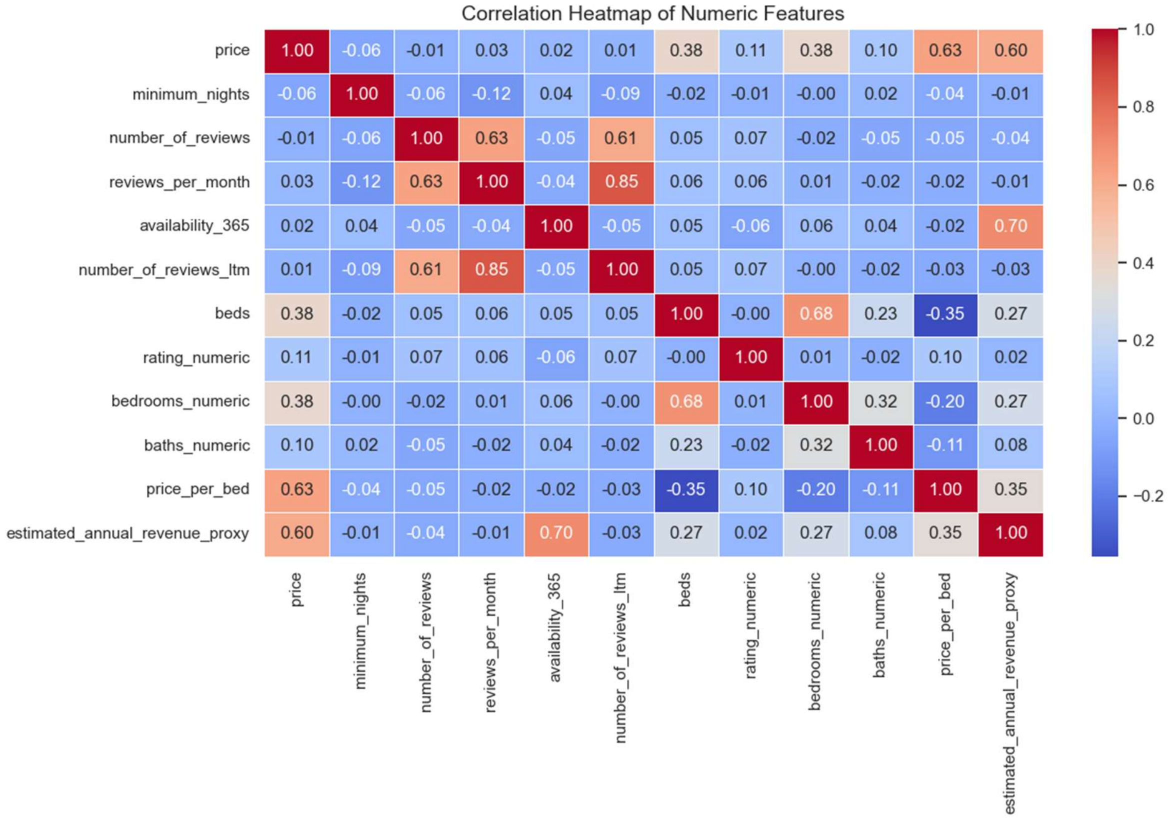

Figure 14: Correlation heatmap of numeric features.

Answer from the analysis:

- High price / high availability listings may represent revenue opportunity if they can convert demand at premium prices.
- Low price / high availability listings may indicate underpriced inventory or weak demand that needs better positioning.
- High price / low availability listings may indicate premium listings with stronger market fit or limited available supply.
- Low price / low availability listings may represent budget-friendly listings that are already competitive or frequently booked.
- The correlation heatmap reinforces that price is not explained by one variable alone; pricing should combine location, room type, beds, baths, availability, and demand signals.

Business takeaway: A quadrant-based opportunity view is useful for revenue management. The next project step could be a pricing recommendation model or demand prediction model using these engineered features.

Final Recommendations

- Benchmark pricing at the borough, neighborhood, and room-type level instead of using overall market averages.
- Prioritize Manhattan and Brooklyn for competitive pricing analysis because they account for nearly three-fourths of focused listings.
- Use neighborhood-level premium clusters such as Greenwich Village, West Village, SoHo, Financial District, Park Slope, and Chelsea for premium market positioning.
- Investigate high-availability listings for potential pricing, policy, or listing quality issues.
- Monitor review velocity and review recency as practical demand indicators alongside price and availability.
- Segment hosts by portfolio size because professional operators and single-listing hosts behave differently.

- Use the engineered features in a future predictive model for price optimization, occupancy forecasting, or host performance scoring.

Skills Demonstrated

- Python data analysis using Pandas and NumPy.
- Data cleaning, duplicate handling, type conversion, missing-value audit, and outlier treatment.
- Feature engineering for business analysis, including availability buckets, stay policy, price per bed, review recency, and host portfolio segmentation.
- Exploratory visualization using Seaborn and Matplotlib.
- Business storytelling through question-driven analysis, chart interpretation, and actionable recommendations.